

OELWEIN, IA, USA

WMO: 725488

Lat: 42.681N

Lon: 91.974W

Elev: 1076

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04955

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-3.8	-16.7	2.3	-8.7	-11.0	3.2	-3.7	31.8	16.9	28.0	17.6	8.7	320	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	90.3	75.8	87.7	74.4	84.4	72.7	79.5	86.2	77.3	83.9	75.2	81.9	12.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	147.9	83.7	75.0	136.5	81.7	72.8	126.7	79.7	43.8	85.5	41.6	82.8	39.4	82.3	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	24.3	20.8	-16.3	92.4	5.4	3.2	-20.1	94.7	-23.3	96.5	-26.3	98.3	-30.2	100.6
			-15.5	82.0	5.0	3.8	-19.1	84.8	-22.0	87.0	-24.8	89.1	-28.4	91.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	19.6	22.3	35.4	48.6	60.3	69.8	72.1	70.0	64.4	50.2	36.8	25.1
	DBStd	21.14	12.56	12.46	11.92	9.73	8.68	6.34	5.61	5.44	7.92	9.66	10.77	11.71
	HDD50	3652	943	777	471	138	13	0	0	0	3	118	414	775
	HDD65	6924	1408	1196	917	497	198	25	9	16	107	466	847	1238
	CDD50	2926	0	1	20	96	333	593	684	619	436	125	17	2
	CDD65	723	0	0	1	5	53	168	228	170	89	9	0	0
	CDH74	6451	0	0	8	94	545	1527	2022	1332	815	91	2	15
	CDH80	1978	0	0	0	22	169	520	646	349	243	19	0	10
	WSAvg	9.8	10.9	10.9	11.4	12.0	11.2	9.2	6.9	6.1	7.7	9.5	10.6	10.8
	PrecAvg	37.6	1.2	1.3	2.1	3.4	5.0	5.9	4.8	4.6	2.8	2.3	2.5	1.3
(15) Precipitation	PrecMax	46.7	2.3	3.4	5.3	8.2	11.3	9.1	10.6	9.4	5.7	5.7	5.3	3.5
	PrecMin	30.2	0.2	0.2	0.1	1.1	1.3	1.1	1.0	0.3	0.4	0.5	0.4	0.4
	PrecStd	5.4	0.6	0.8	1.5	1.6	2.2	2.0	2.5	2.3	1.4	1.4	1.5	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	57.1	73.2	82.4	90.8	93.4	92.8	91.4	90.4	82.0	71.6	62.7
		MCWB	42.2	50.6	59.3	64.0	71.1	73.4	79.7	80.1	72.9	66.1	57.2	58.3
	2%	DB	42.9	48.2	64.0	75.4	84.3	89.6	90.0	87.5	86.2	75.2	63.5	48.4
		MCWB	39.2	44.8	55.1	59.4	66.5	73.0	78.6	76.8	72.0	62.5	54.5	45.0
	5%	DB	37.4	42.6	58.7	70.3	80.8	86.0	86.2	83.8	82.1	71.5	57.3	44.7
		MCWB	35.3	39.0	52.0	56.4	64.5	70.8	76.2	75.2	70.1	61.2	50.8	41.3
	10%	DB	35.7	37.2	52.0	64.8	75.4	82.1	83.7	81.3	79.1	65.9	53.6	38.7
		MCWB	34.1	35.0	46.6	53.6	62.6	69.3	74.8	72.6	67.4	58.4	47.8	36.0
	0.4%	WB	43.8	51.5	61.5	65.5	74.2	80.1	83.7	82.3	77.5	70.9	60.0	58.1
		MCDB	49.8	56.6	72.0	79.2	85.5	87.8	90.4	88.2	85.3	76.4	67.6	61.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.2	44.9	56.9	61.5	70.6	76.9	80.2	78.5	75.1	66.3	55.9	45.7
		MCDB	42.4	48.0	62.5	72.1	80.5	84.4	86.3	85.2	82.1	73.0	61.2	47.2
	5%	WB	36.1	39.1	52.2	58.6	67.9	74.6	77.7	76.3	72.8	62.7	51.7	40.8
		MCDB	38.2	42.0	57.8	67.9	76.3	82.1	83.7	82.2	79.6	68.5	57.3	44.3
	10%	WB	33.5	35.6	46.2	55.1	65.0	71.9	75.4	74.0	70.1	58.4	47.2	36.5
		MCDB	34.9	38.0	51.1	64.0	73.2	79.8	82.3	79.3	76.0	64.6	51.8	38.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	15.4	16.4	20.4	20.2	19.0	18.8	18.8	22.3	19.9	17.0	14.6
		MCDBR	17.3	18.3	25.7	29.5	25.3	23.0	20.2	20.9	24.9	26.1	25.0	19.9
		MCWBR	14.4	14.5	16.5	16.4	11.9	11.4	12.2	12.9	13.5	14.7	16.9	16.6
	5% WB	MCDBR	16.7	17.7	24.2	25.6	21.6	19.4	18.1	18.6	20.4	22.5	23.1	19.1
		MCWBR	14.0	14.4	16.7	16.2	12.4	12.5	12.6	13.0	12.6	15.5	17.0	16.6
(40) Clear-Sky Solar Irradiance	taub		0.278	0.302	0.329	0.370	0.399	0.415	0.429	0.409	0.379	0.335	0.296	0.276
	taud		2.381	2.319	2.373	2.296	2.277	2.272	2.222	2.286	2.339	2.456	2.489	2.421
	Ebn at Noon		275	285	290	285	278	272	267	269	269	270	267	266
	Edh at Noon		25	31	34	40	42	42	44	40	35	27	23	22
(44) All-Sky Solar Radiation	RadAvg		577	902	1210	1473	1751	1934	1964	1704	1405	913	613	452
	RadStd		58	84	128	129	123	162	115	113	105	117	49	37

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page